NTIRE 2026 Image Super-Resolution (×4) Challenge Factsheet Restore Fidelity First, then Generate Details for Perceptual Image Super-Resolution

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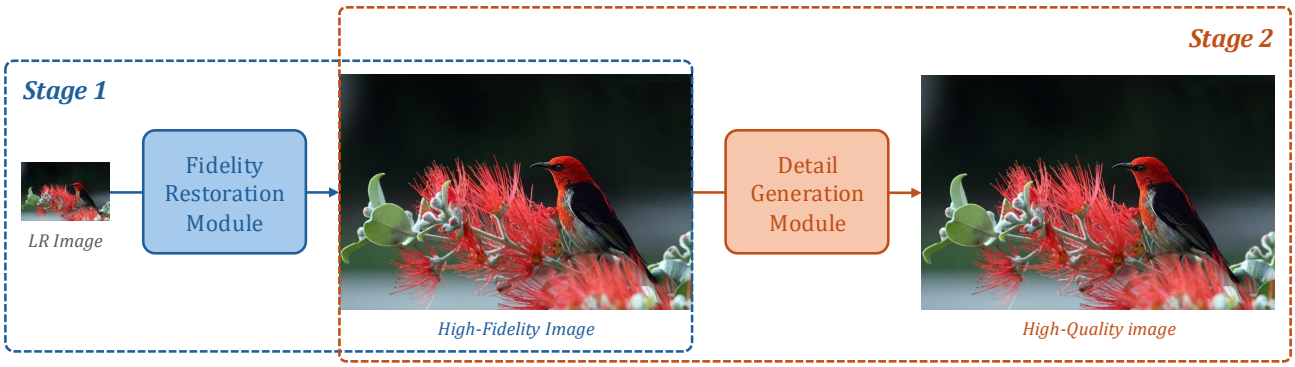

Figure 1. Overview of the two-phase LoRA training strategy used in the detail generation module. Left) In the first phase, we train a LoRA with the flow-matching loss for faithful restoration. Right) In the second phase, we freeze the first-phase LoRA and train an additional LoRA to improve perceptual quality by minimizing LPIPS and maximizing MUSIQ.

1. Factsheet Information

1.1. Team Details

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  - Development (validation): jasonlee, jaeho
  - Test: ciplab
- Link to the codes/executables of the solution(s): github repository

2. Method Details

2.1. Overview

Recent diffusion-based super-resolution (SR) models [5, 16, 24, 28, 31] have shown impressive results by generating realistic details using the rich priors of text-to-image generation models [6, 20, 21]. However, they often struggle to maintain fidelity and structural consistency with respect to the low-resolution (LR) input [2, 10, 18]. To maximize fidelity while still enabling realistic detail generation, we adopt a two-stage pipeline inspired by DiffBIR [16]: restore fidelity first, and then generate realistic details. An overview of our pipeline is shown in Figure 1.

2.2. Stage 1: Fidelity Restoration Module

We considered several candidates for the fidelity restoration module and evaluated them on the DIV2K [1] validation set. Among them, HAT [3] achieved the highest PSNR, specifically with the HAT-L_SRx4_ImageNet-pretrain model. Therefore, we adopt HAT as the first-stage module without further fine-tuning.

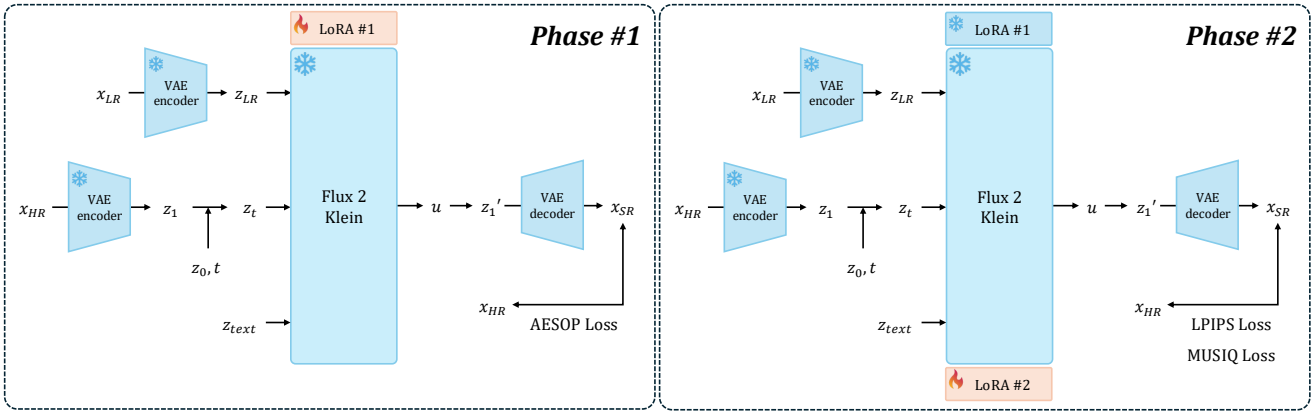

Figure 2. Overview of the training pipeline of our method. Left) We first train a LoRA module with the AESOP loss for faithful restoration. Right) We then train a second LoRA on top of it to further enhance perceptual quality by minimizing LPIPS and maximizing MUSIQ.

2.3. Stage 2: Detail Generation Module

Previous diffusion-based SR models [2, 5, 16, 22, 24, 28, 31, 32] have primarily used text-to-image generation models [6, 20, 21] as their backbones. A key challenge in these approaches is how to inject LR image information while preserving both fidelity and realistic details. Although many methods have been proposed, appropriately incorporating LR information remains far from trivial.

Recently, foundation image editing models that use text instructions [12, 13, 26] have demonstrated impressive performance. Motivated by these advances, we conduct preliminary prompt-based experiments to evaluate their potential for SR. We find that some of these models can generate realistic details while preserving fidelity to a certain extent, even without additional training. This observation suggests that foundation image editing models can serve as a promising starting point for SR, compared with conventional text-to-image generation models.

However, directly applying foundation image editing models to SR presents two major challenges. First, their output fidelity to the LR input remains insufficient. For example, we observe noticeable changes in color tone and structural inconsistencies, such as altered cloud shapes. Second, these models cannot readily handle high-resolution inputs, as they implicitly resize the conditioning image when the input resolution is large. Simply removing this resize operation leads to subpar results. To address these issues, we employ LoRA [9] training for the first challenge and sliding-window inference for the second. Considering our computational budget and preliminary qualitative comparisons, we use FLUX 2 Klein [12] (FLUX 2 Klein base 9B) as the backbone of the detail generation module.

2.3.1. LoRA Training

Various SR models [14, 22, 27, 32, 33] have improved performance by incorporating additional training losses [7, 8, 11, 15, 25, 30, 34]. However, the choice of loss combinations varies widely across studies, and there has been little systematic investigation into which combinations are most effective. To better understand this design space, we conducted extensive experiments with diverse loss combinations [4, 11, 14, 17, 19, 23, 29, 34] and obtained four key observations. 1) Some losses exhibit strong synergy, whereas others do not. 2) Proper weight balancing between losses is essential. 3) Sequential optimization, as also observed in PiSA-SR [22], is more effective than conventional joint optimization. 4) In sequential optimization, the order of losses is crucial.

Motivated by these findings, we adopt a two-phase LoRA training strategy for the detail generation module. In the first phase, we train a LoRA with the AESOP loss [14], which measures the difference between generated and target images in the output space of a pre-trained autoencoder rather than in raw pixel space, to obtain a fidelity-preserving adaptation. In the second phase, we freeze the first-phase LoRA and train an additional LoRA on top of it for perceptual enhancement. Specifically, we optimize a weighted combination of LPIPS [34] and MUSIQ [11]-based objectives, i.e., $L_{LPIPS} + 0.35 \times L_{MUSIQ} = LPIPS + 0.35 \times (1 - MUSIQ)$. The overall training pipeline is illustrated in Figure 2.

2.3.2. Sliding Window Inference

Foundation image editing models cannot readily handle arbitrary high-resolution conditioning inputs. Directly using high-resolution images often leads to suboptimal results. Moreover, simply dividing the image into overlapping patches and stitching the outputs is not well suited for diffusion-based generation, because discrepancies between patches are accumulated over the denoising process. To address this issue, we follow the patch aggregation sampling strategy widely used in diffusion-based SR meth-

ods [16, 24, 31, 32], where overlapping patches are processed independently and their predictions are fused using Gaussian weights at each denoising step.

2.4. Hyperparameters

2.4.1. Train Details

- Dataset: DIV2K train
- Data Augmentation: 1024 x 1024 random crop
- Text prompt: "Use the upscaled condition image as a strict reference. Perform only faithful x4 restoration refinement: reduce blur, ringing, aliasing, and compression artifacts, and recover natural fine details only where supported by the input. Keep the exact same scene layout, object shapes, colors, lighting, and semantics. Improve edge clarity and local texture realism without hallucinating new structures, oversharpening, or changing content."
- LoRA target modules: only to the q, k, v, and out projections in the image stream of the rectified flow transformer
- LoRA rank, alpha: 128, 256
- Iteration: 2000
- Batch size: 2
- Optimizer: AdamW
- Learning rate: 0.000095
- Weight decay: 0.0001

2.4.2. Inference Details

- Text prompt: same text prompt used in train
- Guidance scale: 3.5
- Inference steps: 50
- Patch size: 1024 x 1024
- Patch overlap: 512 x 512

3. Other Details

- Planned submission of a solution(s) description paper at NTIRE 2026 workshop: Perhaps yes.
- General comments and impressions of the NTIRE 2026 challenge: We sincerely appreciate the organizers' hard work for preparing the NTIRE 2026 challenge and providing such a valuable research opportunity for the community.
- What do you expect from a new challenge in image restoration, enhancement and manipulation: Real-world image super-resolution.
- Other comments: For Track 2, which evaluates perceptual quality, the scoring appears to be biased toward NR-IQA metrics. We believe it would be fairer to balance the evaluation more evenly between full-reference IQA metrics and no-reference IQA metrics, for example with a 1:1 ratio: $2 \times ((1 - LPIPS) + (1 - DISTS)) + ((1 - NIQE/10) + ManIQA + MUSIQ/100 + CLIP - IQA)$.

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